

Yaduraj Singh

Software Engineering | Applied AI

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PROFILE

Computer Science & Engineering (AI) student building full-stack applications, AI integrations, and self-hosted infrastructure. Creator of Fleet OS, an open-source container orchestration platform, and coauthor of an IEEE conference paper on multimodal microplastic detection, with contributions to model training and app/backend development.

EDUCATION

Gautam Buddha University

B.Tech, Computer Science & Engineering (AI)

Greater Noida, India

2024 - 2028 (expected)

TECHNICAL SKILLS

Languages	: TypeScript, JavaScript, Python, Go, Swift, C/C++, Dart
Web & APIs	: React, Next.js, Node.js, Fastify, Express, FastAPI, REST APIs
AI & Data	: YOLOv8, faster-whisper, Coqui TTS, LLM APIs, PostgreSQL, Redis, Neo4j, Qdrant, Firebase
Infrastructure	: Docker, BuildKit/Buildx, GitHub Actions, Linux, Nginx, Git
Mobile & IoT	: SwiftUI, Flutter, ESP32-S3, WebRTC, Socket.io, BLE

EXPERIENCE

Lead Software Developer Intern

Gautam Buddha University

2024 (6 months)

Greater Noida, India

- Designed and delivered an institutional ERP, owning database modeling, API contracts, backend implementation, and deployment.
- Built student lifecycle, attendance, fee-ledger, timetable, and faculty/admin modules using **Node.js, Express, React, and JWT-based role access**.
- Iterated with university stakeholders through weekly demos, replacing paper workflows and reducing administrative overhead by an estimated **60%**.

RESEARCH PUBLICATION

Development of a Real-Time Multi Modal Sensor for Field Based Microplastic Detection

S. Upadhayay, V. Bhadaurya, **Yaduraj Singh**, V. Dwivedi, Y. Singh, and Sharmishtha.

2026 2nd International Conference on Cognitive Computing in Engineering, Communications, Sciences and Biomedical Health Informatics (IC3ECSBHI) , IEEE, pp. 736–741, 2026.

[IEEE Xplore](#) | [DOI: 10.1109/IC3ECSBHI67834.2026.11469066](https://doi.org/10.1109/IC3ECSBHI67834.2026.11469066)

- Contributed to **model training and app/backend development** for the multimodal microplastic-detection research system.
- The published system combines **YOLOv8-based microscopy**, impedance measurements, and multispectral optical sensing for microplastic analysis in water samples.

Fleet OS | Open-Source Container Orchestration

[GitHub](#) | [Website / Docs](#)

Go, TypeScript, Fastify, PostgreSQL, Redis, Docker BuildKit/Buildx, React

- Built a control plane, CLI, and **outbound-only Go agents** to coordinate Linux containers across ARM/x86 machines and VPSs, including worker nodes behind NAT.
- Implemented placement using node capacity, platform compatibility, and service constraints; added health-gated rollouts and rescheduling for eligible stateless services while keeping stateful workloads pinned.
- Implemented delegated **BuildKit** jobs on opted-in nodes, scoped registry credentials, and immutable image-digest deployments. Validated the build/deploy path on Apple Silicon; broader release testing remains in progress.

Aarogya Setu | Hospital Management SaaS

[GitHub](#) | [Live Site](#)

Next.js, TypeScript, Prisma, PostgreSQL, Redis, Docker

- Built a multi-tenant platform for patient records, OPD/IPD queues, appointments, billing, pharmacy, and diagnostic workflows, with **six-tier role-based access control**.
- Containerized services and configured GitHub Actions deployment to a Linux server with Nginx and TLS; used PostgreSQL for application data and Redis for queue/session state.

Tollgate | LLM API Cost & Usage Observability

[Live Site](#)

LLM APIs, OpenAI-compatible proxy, Anthropic, OpenRouter

- Built an API proxy that attributes LLM usage and spending to application features through a configurable base URL.
- Added exact-match prompt caching, gateway-enforced spending caps, and runaway-agent alerts to help control repeated requests and unexpected usage.

SecondMind / CortX | Embedded Voice Assistant

Python, FastAPI, ESP32-S3, Flutter, Neo4j, Qdrant

- Built a voice pipeline connecting **faster-whisper**, a locally hosted LLM, and **Coqui TTS**, with a Flutter interface and ESP32-S3 audio capture.
- Implemented voice activity detection and Opus audio encoding; integrated Neo4j conversation graphs and Qdrant semantic memory.

MuhDikhai | Peer-to-Peer Video Chat

[GitHub](#)

Node.js, TypeScript, WebRTC, Socket.io

- Implemented WebRTC offer/answer exchange, ICE negotiation, and custom Socket.io signaling, with matchmaking, disconnect handling, and reconnection logic.

CineVerse | Film Discovery & Social Tracking

[Live Site](#)

Next.js, Firebase, TMDB API, Nginx, PM2

- Built watchlists, ratings, reviews, and discovery feeds using TMDB data and Firebase; deployed the server-rendered application on Linux with Nginx and PM2.